

u3-waves_and_quantum_optics

n01-interference

Interference

- when two or more waves are superimposed the resultant amplitude is given by the algebraic sum of amplitudes of individual waves.
- Since the intensity is directly proportional to the square of amplitude, the modification in the intensity in the region of superposition is known as interference.
- when the two waves are superimposed in such a way that the resultant is sum of the individuals, the interference is called constructive interference. (fig(a))
- If the resultant is difference of the individuals as shown in fig(b) the interference is destructive interference.

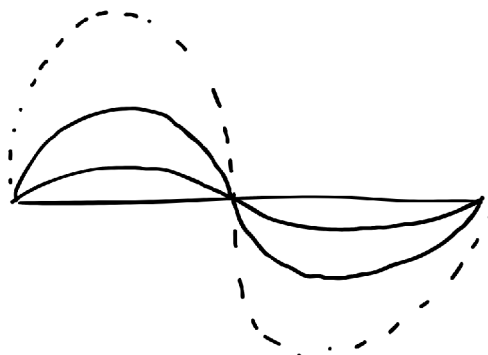


fig (a)

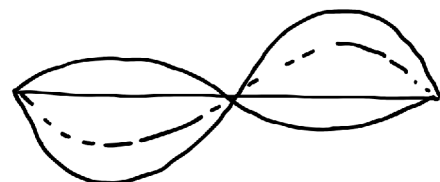
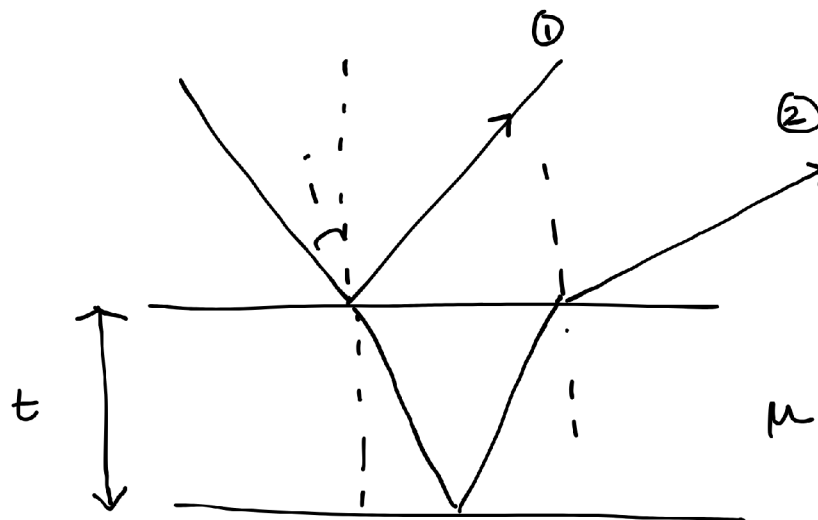


fig (b)

Conditions for sustained Interference.

- The two light sources must emit coherent waves.
- The two light sources must emit continuous waves with same frequency and wavelength.
- The distance between the sources must be small.
- The distance between the screen and two sources must be large.
- The Amplitudes of superimposed waves must be equal or nearly equal.
- The background must be dark to observe the bright fringes.

nO2-thin_film_interference



- when the light ray is incident on a thin film of thickness ' t ', some of the light gets reflected from the upper surface of the film and some will refract into the film and then get reflected from the lower surface of the film.
- The path difference between the two reflected rays is given by

$$\Delta x = 2\mu t \cos r + \frac{\lambda}{2}$$

Where μ is the refractive index of the medium.

- For bright fringe

$$\text{path difference} = n\lambda$$

$$2\mu t \cos r + \frac{\lambda}{2} = n\lambda$$

$$2\mu t \cos r = n\lambda - \frac{\lambda}{2}$$

$$2\mu t \cos r = \frac{(2n - 1)\lambda}{2}$$

- For Dark fringe

- path difference = $(2n + 1)\frac{\lambda}{2}$

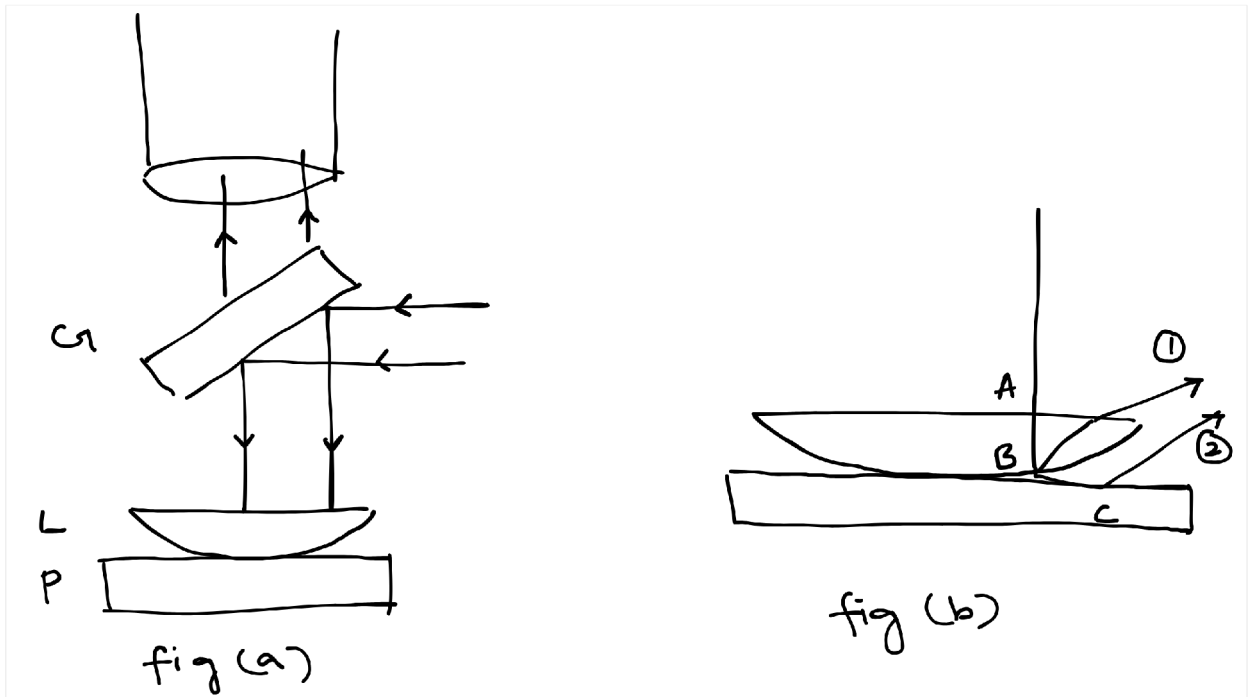
- $2\mu t \cos r + \frac{\lambda}{2} = (2n + 1)\frac{\lambda}{2}$

- $2\mu t \cos r = (2n + 1)\frac{\lambda}{2} - \frac{\lambda}{2}$

$$2\mu t \cos r = n\lambda$$

n03-newton's_rings

- Newton's rings is the best example for the interference observed from a thin film of non-uniform thickness. when a planoconvex lens is placed on plane glass plate, a non-uniform air film is formed between the two.
- The thickness of the air film at point of contact is zero.
- If a monochromatic light is incident normally on the arrangement and viewed the reflected light gives alternate bright and dark rings concentric about the point of contact.
- Name the rings are Newton's rings as he discovered them.



The experimental arrangement.

- shown in fig (a)
 - G - glass plate
 - L - Planoconvex lens
 - P - Plane glass plate
 - M - Microscope
- Let us consider an incident light AB falls normally on thin air film at B the light ray gets reflected as ray (1)
- And some of light transmitted along BC and reflected at C (air-glass boundary) as ray (2).
- Now the path difference between ray (1) and ray (2) is given by

$$\Delta x = 2\mu t \cos r + \frac{\lambda}{2}$$

- For air film $\mu = 1$ and for normal incidence $r = 0$

$$\Delta x = 2(1)t \cos(0) + \frac{\lambda}{2}$$

$$\Delta x = 2t + \frac{\lambda}{2} \dots (i)$$

- The formation of bright and dark film depends on the thickness of the film.
- At point of contact

- $t = 0$

- $\Delta x = \frac{\lambda}{2}$

- $\Delta\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi \implies$ out of phase (Dark fringe)

- For bright ring

- $\Delta x = n\lambda$

- $2t + \frac{\lambda}{2} = n\lambda$

- $2t = n\lambda - \frac{\lambda}{2}$

- $2t = \frac{(2n-1)\lambda}{2} \dots (ii)$

- For Dark Ring

- $\Delta x = (2n+1)\frac{\lambda}{2}$

- $2t + \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2}$

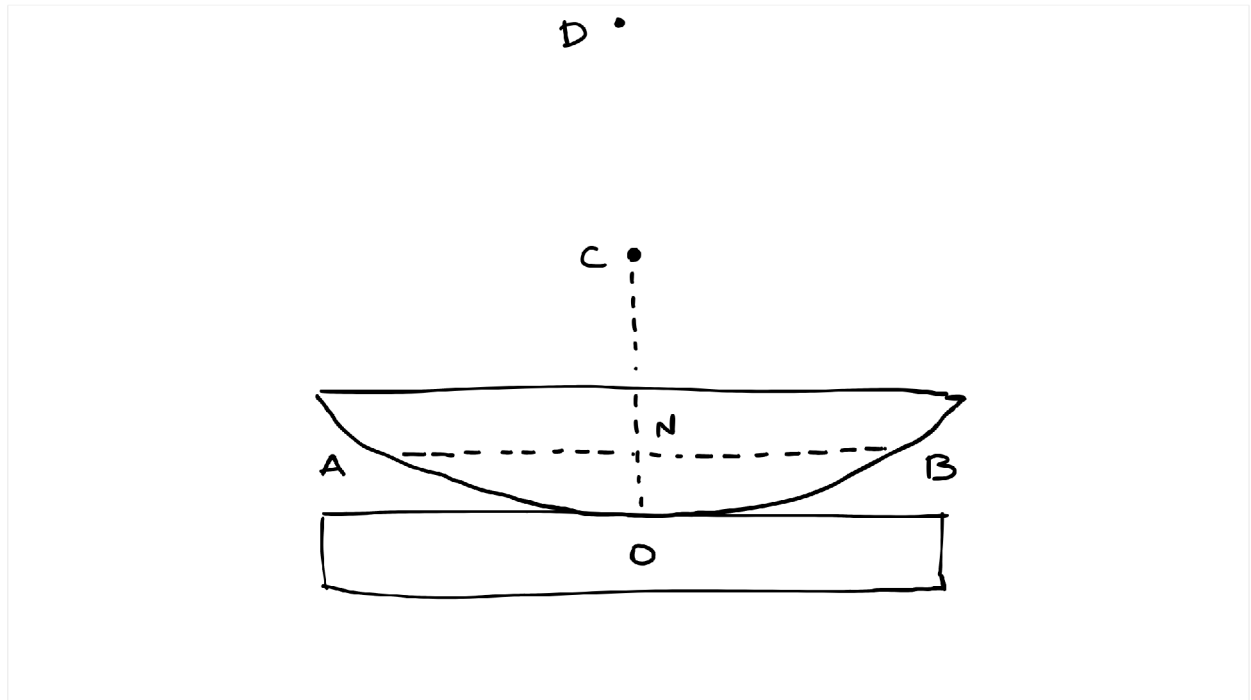
- $2t = (2n+1)\frac{\lambda}{2} - \frac{\lambda}{2}$

- $2t = \frac{2n\lambda}{2}$

- $2t = n\lambda \dots (iii)$

- To find the diameters of bright and dark rings. Let L be the planoconvex lens of large radius of curvature is placed on a plate plane glass plate P, Such that a convex surface touches the plate plane surface at O.
- Let us also consider the convex surface is part of a spherical surface with C as centre, let R be the radius of curvature of lens and r is the radius of

the Newton's rings.



- From geometry, for a point t on the ring,

$$NA \times NB = NO \times ND$$

$$r \times r = t(2R - t)$$

$$r^2 = 2Rt - t^2$$

- t^2 is neglected because t is small

$$r^2 \approx 2Rt \dots (iv)$$

- To find the diameter of bright ring
 - From (ii) and (iv)

$$2t = \frac{r^2}{R}$$

$$\frac{r^2}{R} = \frac{(2n - 1)\lambda}{2}$$

$$2r^2 = R(2n - 1)\lambda$$

- If D_n is a diameter of bright ring then $r = \frac{D_n}{2}$

$$\frac{D_n^2}{4} = R \frac{(2n - 1)\lambda}{2}$$

$$D_n^2 = 2\lambda R(2n - 1)$$

$$D_n = \sqrt{2\lambda R(2n - 1)} \dots (v)$$

- To find the diameter of Dark ring
 - From (iii) and (iv)

$$2t = \frac{r^2}{R}$$

$$\frac{r^2}{R} = n\lambda$$

$$r^2 = n\lambda R$$

- If D_m is a diameter of dark ring then $r = \frac{D_m}{2}$

$$\frac{D_m^2}{4} = n\lambda R$$

$$D_m^2 = 4n\lambda R \dots (vi)$$

$$D_m = 2\sqrt{n\lambda R}$$

- This means that the rings are more closely spaced when we move away from the point of contact.

n04-determination_of_wavelength_of_sodium_light

- Newton's rings are formed by placing the convex lens L on plane glass plate P. The diameter of different dark ring can be found as follows.
- From (vi) for m^{th} ring and n^{th} ring ($m > n$)

$$D_m^2 = 4m\lambda R \dots (i)$$

$$D_n^2 = 4n\lambda R \dots (ii)$$

- Subtracting (ii) from (i)

$$D_m^2 - D_n^2 = 4m\lambda R - 4n\lambda R$$

$$D_m^2 - D_n^2 = 4(m - n)\lambda R$$

$$\lambda = \frac{D_m^2 - D_n^2}{4(m - n)R}$$

where R is the radius of curvature of L which is determined by using spherometer.

- The term $\frac{D_m^2 - D_n^2}{m - n}$ is the slope of the graph plotted by taking number of rings on x-axis and square of the diameter on y-axis.

n05-

determination_of_refractive_index_of_unknown_liquid

- Newton's rings should be done when air film is between L and P

$$D_m^2 - D_n^2 = 4(m - n)\lambda R \dots (i)$$

- When the liquid film is between L and P (where μ is the refractive index of liquid) the path difference is $2\mu t \cos r$.

$$\text{iameter of dark ring in liuid } D_m'^2 = \frac{4m\lambda R}{\mu}$$

$$D_m'^2 - D_n'^2 = \frac{4(m-n)\lambda R}{\mu} \dots (ii)$$

- Dividing (i) by (ii)

$$\frac{D_m^2 - D_n^2}{D_m'^2 - D_n'^2} = \frac{4(m-n)\lambda R}{4(m-n)\lambda R/\mu}$$

$$\frac{D_m^2 - D_n^2}{D_m'^2 - D_n'^2} = \mu$$

$$\mu = \frac{D_m^2 - D_n^2}{D_m'^2 - D_n'^2}$$

n06-diffraction

- When light falls on an obstacle of size comparable to that of the wavelength of light used, bending takes place around the edges or corners of the obstacles and light enters into the geometrical shadow.
- The phenomenon of bending of light is called diffraction.
- Diffraction is broadly divided into two types namely Fraunhofer and Fresnel diffraction.

n07-fraunhofer_diffraction

1. For this diffraction to occur the light source and screen must be placed at infinite distance from obstacle.
2. 2 converging lens are required.
3. Study of their diffraction is easy.
4. Incident wavefront is a plane wave.
5. Studied in any direction of propagation.

n08-fresnel_diffraction

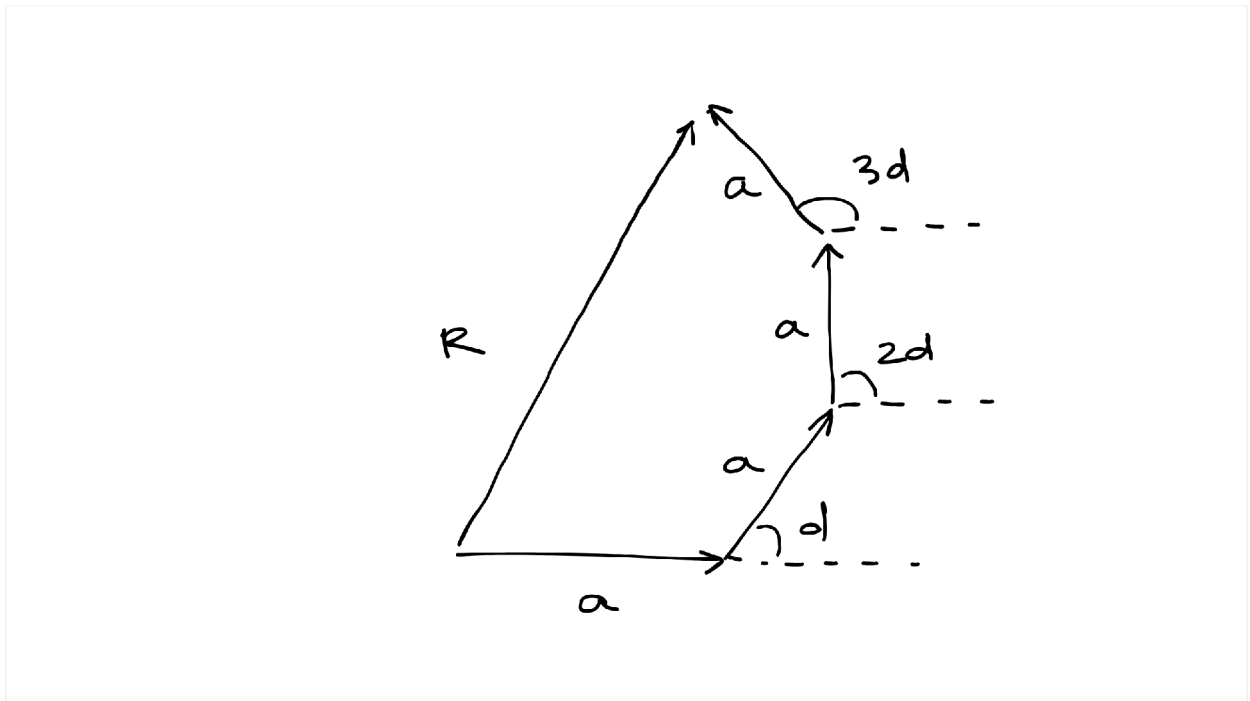
1. For this diffraction to occur distance between light source and screen must be finite.
2. No lens are required.
3. The study of is complicated.
4. Incident wavefront is either circular or spherical.
5. Studied only in the direction of propagation.

nO9-fraunhofer_diffraction_at_single_slit

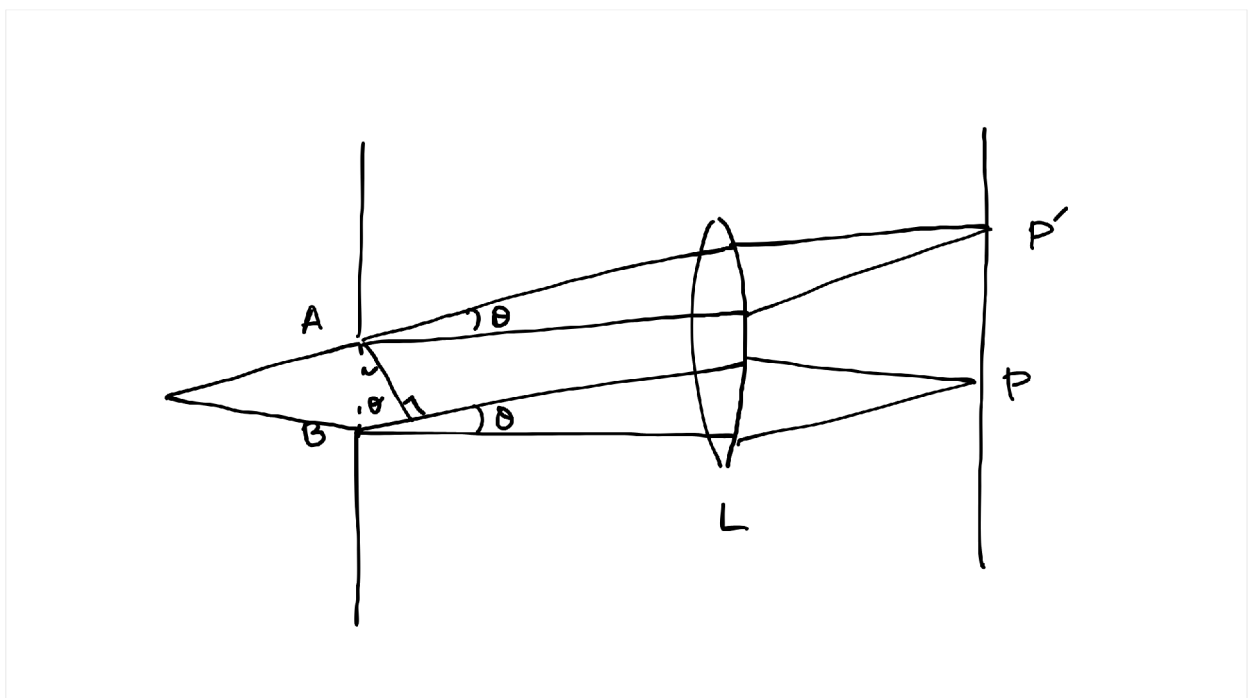
Resultant of n simple harmonic waves

- If there are n simple harmonic vibrations having same amplitude (a), and common phase difference (d) between two successive waves then the resultant amplitude is given by the vector addition which is obtained as

$$R = \frac{a \sin\left(\frac{nd}{2}\right)}{\sin\left(\frac{d}{2}\right)}$$



- Let a monochromatic light propagation normally is incident and a slit AB of width e .



- The diffracted light to the slit is focused by means of a converging lens L and a screen placed in the focal plane of the lens.
- According to Huygen's theory, **each point in the slit acts as a source of secondary wavelets and spreads the secondary wavelets towards the screen.**

The waves which travel in the direction of OP are focused at P on the screen.

- Thus **the intensity at P is maximum.**
- The wavelets which are making an angle θ with the original direction are brought to focus at P' .
- **The intensity may be maximum or minimum on P' , which depends on the path difference.**
- To find intensity at P' , let us draw a normal from A to B .

$$\sin \theta = \frac{BC}{AB} \quad \text{from diagram}$$

$$\Delta x = BC = AB \sin \theta$$

$$\Delta x = e \sin \theta \quad \dots (i)$$

$$\Delta \phi = \frac{2\pi}{\lambda} \cdot e \sin \theta \quad \dots (ii)$$

- Let the slit be divided into n equal parts such that each part sends the secondary waves of equal amplitude A and

$$\Delta \phi = \frac{1}{n} \left[\frac{2\pi}{\lambda} e \sin \theta \right] = d \quad (\text{sa})$$

- By vector addition method

$$R = \frac{a \sin \left(\frac{nd}{2} \right)}{\sin \left(\frac{d}{2} \right)}$$

$$R = \frac{a \sin \left(\frac{n}{2} \frac{2\pi e \sin \theta}{n\lambda} \right)}{\sin \left(\frac{1}{2} \frac{2\pi e \sin \theta}{n\lambda} \right)}$$

$$R = \frac{a \sin \left(\frac{\pi e \sin \theta}{\lambda} \right)}{\sin \left(\frac{\pi e \sin \theta}{n\lambda} \right)}$$

Let

$$= \frac{\pi e \sin \theta}{\lambda}$$

$$R = \frac{a \sin}{\sin \left(\frac{1}{n} \right)}$$

since $\frac{1}{n}$ is small,

$$\sin\left(\frac{a}{n}\right) \approx \frac{a}{n}$$

$$R = \frac{a \sin}{\left(\frac{a}{n}\right)}$$

$$R = \frac{na \sin}{n}$$

Let $A = na$. Here A is the amplitude when all the secondary waves are in phase.

$$R = \frac{A \sin}{n} \quad \dots (iii)$$

Condition for principal maximum

The principal maximum can be obtained by expanding $\sin$ in $\dots (iii)$

$$R = \frac{A}{n} \left[-\frac{3}{3} + \frac{5}{5} - \dots \right]$$

$$R = A \left[1 - \frac{2}{3} + \frac{4}{5} - \dots \right]$$

so in above expression if all negative terms become zero, then R will be maximum and that will only happen when $\theta = 0$.

$$\frac{\pi e \sin \theta}{\lambda} = 0$$

$$\sin \theta = 0$$

$$\theta = 0$$

The central maxima occurs when $\theta = 0$ and $I_{\max} = A^2 = R^2$.

Condition for minimum

The minimum intensity can be observed when $\sin \theta = 0$.

$$= \pm m\pi, \quad m = 1, 2, 3 \dots$$

$$\frac{\pi e \sin \theta}{\lambda} = \pm m\pi$$

$$e \sin \theta = \pm m\lambda$$

The position of secondary maxima by differentiating the intensity w.r.t θ and equating it to zero.

$$\frac{dI}{d\theta} = 0$$

$$\frac{d}{d\theta} \left[\left(\frac{A \sin}{n} \right)^2 \right] = 0$$

$$2A^2 \left(\frac{\sin}{2} \right) \left[\frac{\cos - \sin}{2} \right] = 0$$

$$\cos - \sin = 0$$

$$\cos = \sin$$

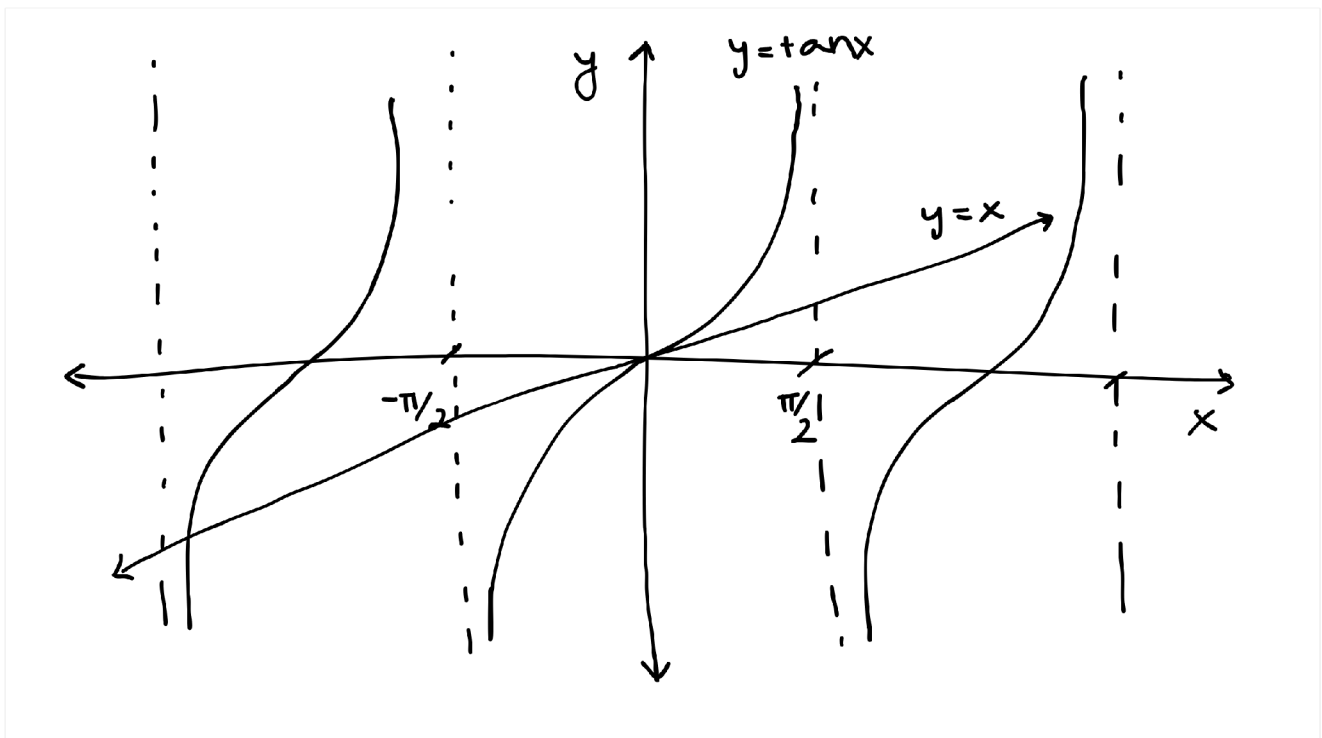
$$= \tan$$

- this is only possible if

$$\sin = 0 \implies \text{gives position of minima}$$

$$\cos - \sin = 0$$

- this equation can be solved by plotting the graph $y = \sin x$ and $y = \tan x$.



- From the above graph, the points of intersections are

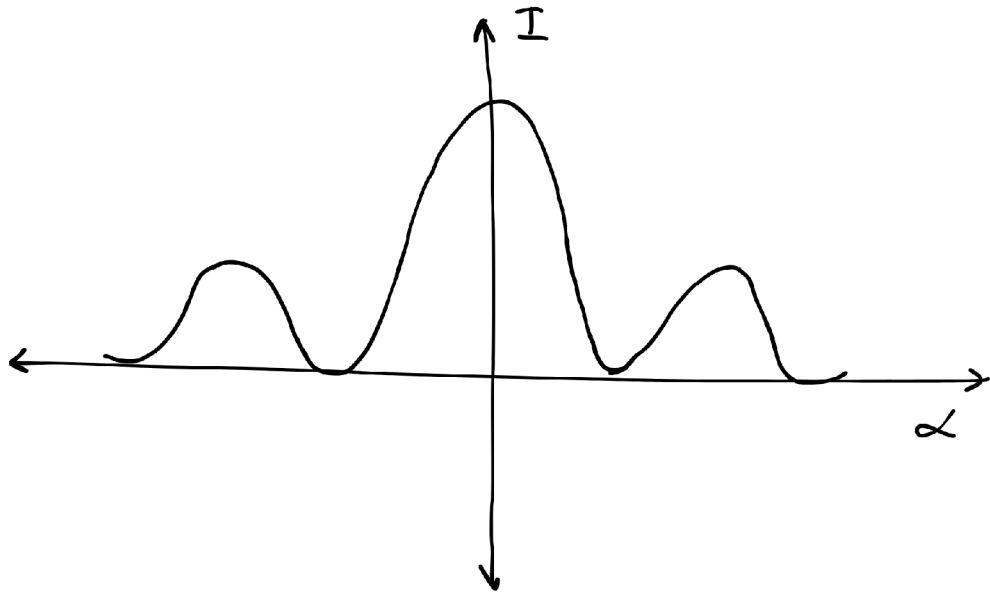
$$= 0, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}$$

$$= 0, I_0 = R^2 = A^2$$

$$= \pm \frac{3\pi}{2} \implies I_1 = A^2 \frac{\sin^2(3\pi/2)}{(3\pi/2)^2} \approx \frac{A^2}{22}$$

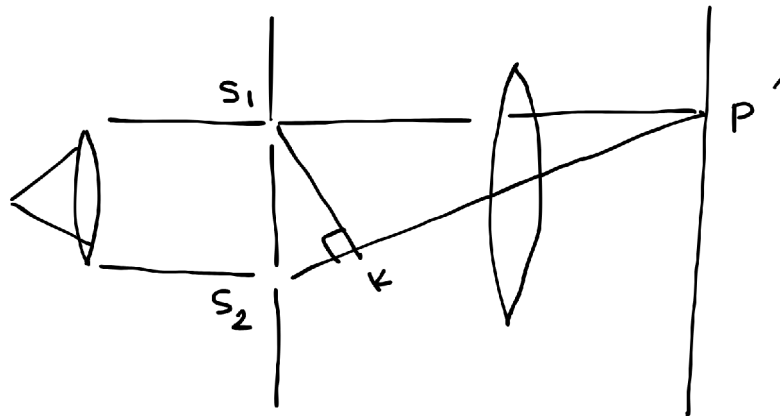
$$= \pm \frac{5\pi}{2} \implies I_2 \approx \frac{I_0}{62}$$

Intensity distribution due to single slit



Diffraction due to double slit

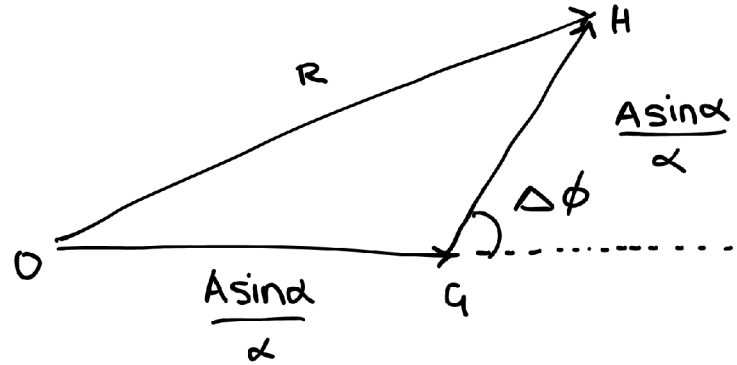
- Let S be the monochromatic source and consider 2 slits of equal width ' e ' which are separated by the distance ' d '



- The amplitude of wavelets from S_1 is $\frac{A \sin}{\lambda}$
- To know the phase difference between the waves emerging from S_1 and S_2 , let us draw a normal from S_1K on S_2K .
- The path difference

$$\Delta x = S_2K = (e + d) \sin \theta$$

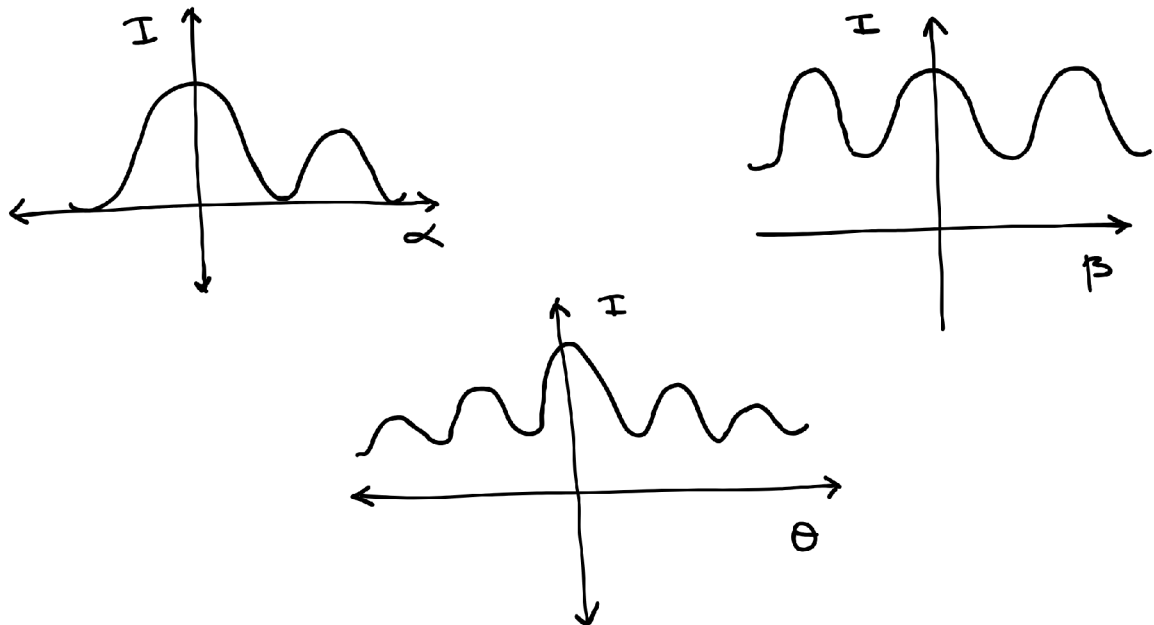
$$\Delta \phi = \frac{2\pi}{\lambda} (e + d) \sin \theta$$



- The resultant amplitude is given by the vector addition method.

$$\begin{aligned}
 R^2 &= (OA)^2 + (AH)^2 + 2(OA)(AH) \cos \Delta\phi \\
 &= \left(\frac{A \sin}{\alpha} \right)^2 + \left(\frac{A \sin}{\alpha} \right)^2 + 2 \left(\frac{A \sin}{\alpha} \right)^2 \cos \Delta\phi \\
 &= 2 \left(\frac{A \sin}{\alpha} \right)^2 [1 + \cos] \\
 &= 2 \left(\frac{A \sin}{\alpha} \right)^2 \left[2 \cos^2 \left(\frac{\Delta\phi}{2} \right) \right] \\
 &= 2 \left(\frac{A \sin}{\alpha} \right)^2 \left[2 \cos^2 \left(\frac{\pi(e+d) \sin \theta}{\lambda} \right) \right] \\
 R^2 &= 4 \left(\frac{A \sin}{\alpha} \right)^2 \cos^2 \\
 &= \frac{\pi(e+d) \sin \theta}{\lambda} \\
 I &= 4 \left(\frac{A^2 \sin^2}{2} \right) \cos^2
 \end{aligned}$$

- In the above expression, the term $\frac{A^2 \sin^2}{2}$ gives the effect of diffraction due to single slit.
- The term $\cos^2$ gives the effect of interference.
- Therefore the resultant intensity is due to the combined effect of both interference and diffraction.



n10-diffraction

- An arrangement of which consists of equidistant parallel lines of same size is known as grating
- It may be constructed by drawing lines on a transparent glass plate.
- The rules are opaque to light and the area between the lines is transparent to light
- If e is the width of the line and d is the width of the slit. then $(e+d)$ is called grating element.
- If N is the number of lines for each in per inch of the grating, then

$$N(e + d) = 1 = 2.54cm$$

$$e + d = \frac{2.54cm}{N}$$

n11-

superposition_of_two_simple_harmonic_vibrations_i n_1d

- Consider two simple harmonic vibrations having equal frequency ω and different amplitudes a_1 and a_2 , different phases ϕ_1 and ϕ_2 travelling in x-direction

$$x_1 = a_1 \cos(\omega t + \phi_1) \dots (i)$$

$$x_2 = a_2 \cos(\omega t + \phi_2) \dots (ii)$$

- According to the principal of superposition, the resultant displacement is given by

$$x = x_1 + x_2$$

$$= a_1 \cos(\omega t + \phi_1) + a_2 \cos(\omega t + \phi_2)$$

$$x = a_1 \cos \omega t \cos \phi_1 - a_1 \sin \omega t \sin \phi_1 + a_2 \cos \omega t \cos \phi_2 - a_2 \sin \omega t \sin \phi_2$$

$$x = \cos \omega t (a_1 \cos \phi_1 + a_2 \cos \phi_2) - \sin \omega t (a_1 \sin \phi_1 + a_2 \sin \phi_2)$$

- Put $A \cos \phi = a_1 \cos \phi_1 + a_2 \cos \phi_2$
- $A \sin \phi = a_1 \sin \phi_1 + a_2 \sin \phi_2$

$$= A \cos \omega t \cos \phi - A \sin \omega t \sin \phi$$

$$x = A \cos(\omega t + \phi)$$

- therefore the resultant of 2 collinear simple harmonic oscillations having equal frequency is again a simple harmonic oscillation with different amplitude 'A' and different phase ϕ
- According to the principal of superposition

$$x = A \cos(\omega t + \phi)$$

where $A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$

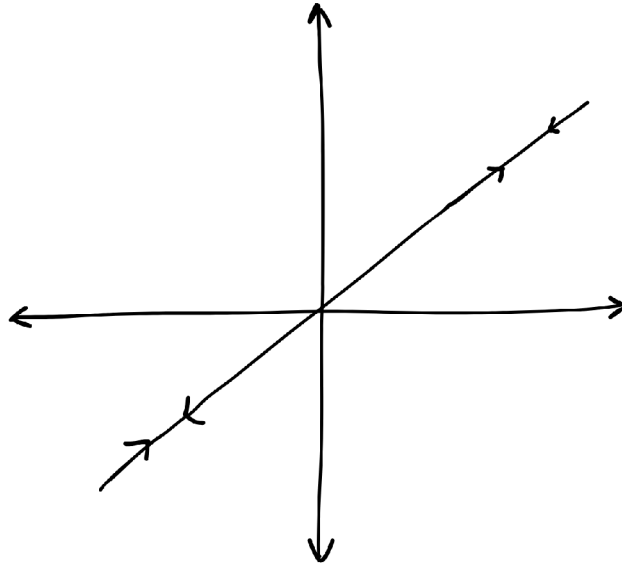
$$\tan \phi = \frac{a_1 \sin \phi_1 + a_2 \sin \phi_2}{a_1 \cos \phi_1 + a_2 \cos \phi_2}$$

n12-lissajous_figures

- When a particle is influenced simultaneously by 2 simple harmonic vibration which are travelling at right angle to each other, the resultant phase of the particle is called Lissajous figure.
- The shape of the Lissajous figure depends mainly on 3 factors.
 - Amplitude of 2 vibrations
 - frequency of 2 vibrations
 - Phase difference
- Discuss the different cases

$$\left. \begin{aligned} x &= A_1 \cos \omega t \\ y &= A_2 \cos(\omega t + \phi) \end{aligned} \right\} \dots (1)$$

case 1: $\phi = 0$

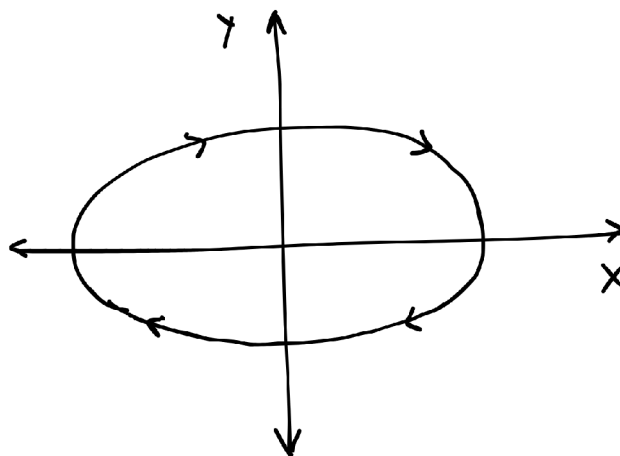


$$y = A_2 \cos \omega t$$

$$\cos \omega t = \frac{x}{A_1}$$

$$y = \left(\frac{A_2}{A_1} \right) x.$$

case 2: $\phi = \pi/2$



$$y = A_2 \cos(\omega t + \pi/2)$$

$$y = -A_2 \sin \omega t.$$

from (1) $\cos \omega t = \frac{x}{A_1}$

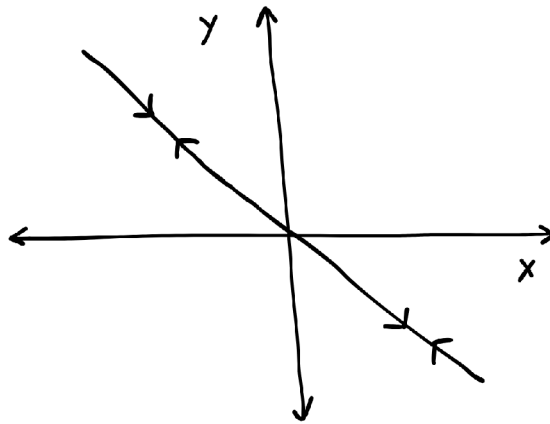
$$\sin \omega t = \sqrt{1 - \frac{x^2}{A_1^2}}$$

$$y^2 = A_2^2 \sin^2 \omega t$$

$$y^2 = A_2^2 \left(1 - \frac{x^2}{A_1^2}\right)$$

$$\frac{y^2}{A_2^2} + \frac{x^2}{A_1^2} = 1$$

case 3: $\phi = \pi$



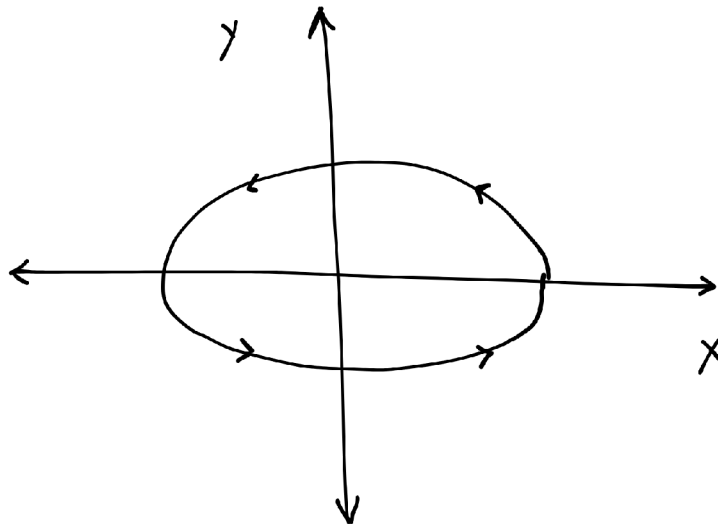
$$y = A_2 \cos(\omega t + \pi)$$

$$y = -A_2 \cos \omega t$$

$$\cos \omega t = \frac{x}{A_1}$$

$$y = -A_2 \frac{x}{A_1}$$

case 4: $\phi = \frac{3\pi}{2}$



$$y = A_2 \cos\left(\omega t + \frac{3\pi}{2}\right)$$

$$y = -A_2 \sin \omega t$$

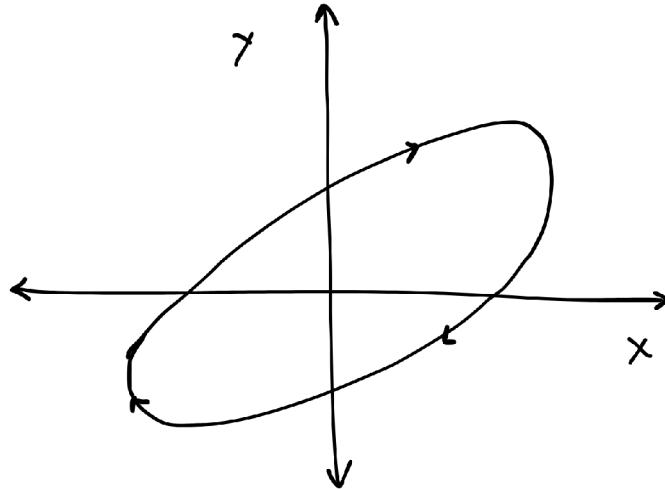
$$\cos \omega t = \frac{x}{A_1}$$

$$\sin \omega t = \sqrt{1 - \frac{x^2}{A_1^2}}$$

$$y^2 = A_2^2 \sin^2 \omega t$$

$$\frac{y^2}{A_2^2} + \frac{x^2}{A_1^2} = 1$$

case 5: $\phi = \pi/4$



$$y = A_2 \cos(\omega t + \pi/4)$$

$$y = A_2 \left(\frac{\cos \omega t}{\sqrt{2}} - \frac{\sin \omega t}{\sqrt{2}} \right)$$

$$y = \frac{A_2}{\sqrt{2}} (\cos \omega t - \sin \omega t)$$

$$\cos \omega t = \frac{x}{A_1} \quad \sin \omega t = \sqrt{1 - \frac{x^2}{A_1^2}}$$

$$y^2 = \frac{A_2^2}{2} (\cos^2 \omega t + \sin^2 \omega t - 2 \sin \omega t \cos \omega t)$$

$$y^2 = \frac{A_2^2}{2} (1 - 2 \sin \omega t \cos \omega t)$$

$$y^2 = \frac{A_2^2}{2} 1 - 2 \frac{x \sqrt{A_1^2 - x^2}}{A_1^2}$$

$$y^2 = \frac{A_2^2}{2A_1^2} (A_1^2 - 2x \sqrt{A_1^2 - x^2})$$

$$y = \frac{A_2}{\sqrt{2}A_1} \sqrt{A_1^2 - 2x \sqrt{A_1^2 - x^2}}$$

$$\frac{2y^2}{A_2^2} + \frac{2x^2}{A_1^2} - \frac{2\sqrt{2}}{A_1 A_2} xy = 1$$

[Diagram of Oblique Ellipse]